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HOST MICROBE INTERACTIONS

Bartonella, Rodents, Fleas and Ticks: a Molecular Field Study on Host-Vector-Pathogen Associations in Saxony, Eastern Germany

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Abstract Bartonellae cause zoonotic diseases and are transmitted by arthropods. Rodents are reservoirs for most *Bartonella* spp. As the knowledge about *Bartonella* in rodents and their parasitizing ectoparasites is scarce in Germany, this study's objectives were to investigate *Bartonella* spp. in small mammals and in their ectoparasites. A total of 79 small mammals (seven species) were captured and their ectoparasites collected at seven sites around Leipzig, Saxony, Germany, in 2010 and 2011. Altogether, 79 spleen samples, 135 fleas (five species) and 365 ticks (three species) were investigated for *Bartonella* spp. by PCR targeting the ITS 16S–23S rRNA region. In total, 52 (65.8 %) small mammals, 73 (54.1 %) fleas and 51 (16.3 %) ticks were positive for *Bartonella* spp. Most small mammals were positive for uncultured *Bartonella* sp. ($n=29$) followed by *Bartonella grahamii* ($n=12$), *Bartonella taylorii* ($n=8$) and *Bartonella* sp. N40 ($n=3$). Likewise, most fleas were positive for uncultured *Bartonella* sp. ($n=45$) followed by *B. grahamii* ($n=14$), *B. taylorii* ($n=8$), *B.* sp. N40 ($n=5$) and *Bartonella elizabethae* ($n=2$). Most ticks were positive for *B.* sp. ($n=19$) followed by *B. grahamii* ($n=10$), *Bartonella chomelii* ($n=3$), *B. taylorii* ($n=2$) and *B.* sp. N40 ($n=1$). This study's results suggest that rodents and fleas may be reservoirs and vectors, respectively. Zoonotic *B. grahamii* and *B. elizabethae* were found in rodents and their fleas. Therefore, humans may contract *Bartonella* infection by contact to wild rodents. Ticks seem of minor importance in transmitting *Bartonella* spp. found in fleas and rodents. However, ticks might be vectors of *B. chomelii*.

Keywords *Apodemus flavicollis* · *Myodes glareolus* · *Ctenophthalmus nobilis* · *Bartonella* spp. · *Ixodes ricinus*

Introduction

Bartonella spp. are facultative intracellular alphaproteobacteria (order Rhizobiales, family Bartonellaceae) and the causative agents of bartonellosis which may result in severe clinical symptoms in humans and domesticated animals worldwide [39]. Bartonellosis, depending on the infecting *Bartonella* species, may cause chronical (e.g. verruga peruana) and acute (e.g. cat scratch fever) infections displayed by a variety of clinical symptoms [26, 40]. *Bartonella quintana*, *Bartonella bacilliformis* and *Bartonella henselae* are the most common causative agents and are carried by humans and cats (*B. henselae*) respectively [39]. However, there is an increasing number of clinical reports in humans caused by other bartonellae such as *Bartonella elizabethae* and *Bartonella grahamii* which are harboured by rodents [53]. While varying prevalences of bartonellae were found in wild ungulates, carnivores and domesticated animals [9, 12, 19, 32], the highest prevalence and diversity of *Bartonella* spp. were reported in rodents which are considered to be main reservoirs. To date, about 30 *Bartonella* species are

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known of which ten are harboured by rodents and five of the latter are known to be of zoonotic potential (*B. grahamii*, *B. elizabethae*, *Bartonella vinsonii* subsp. *arupensis*, *Bartonella washoensis* and *Bartonella tamiae*) [14]. High prevalences of *Bartonella* spp. were reported in rodents from Asian [51] (21 %), African [29] (34 %) and American countries [22] (42 %). In Europe, the prevalence of *Bartonella* spp. in rodents is ranging from 13.7 to 57.5 % in studies from Poland [35, 54], Slovenia [21], Spain [13], France [8] and the UK [49].

Bartonella spp. are mainly transmitted by haematophagous arthropod vectors such as fleas, lice, sandflies and ticks [2, 4, 14, 44]. Other vectors such as the woodlouse hunter spider (*Dysdera crocata*) have been suggested as vectors as well [27]. Amongst ectoparasites infesting rodents, fleas are expected to be main vectors for *Bartonella* spp. [3, 52]. *Ctenophthalmus nobilis* also known as *Ctenophthalmus agyrtes*, the most common flea species parasitizing rodents in Central Europe, is known to transmit *B. grahamii* and *B. taylorii* to *Myodes glareolus* [5]. In European countries such as Poland [38], Italy [42], the Netherlands [44] and France [15], *Bartonella* spp. were detected in *Ixodes ricinus*, the most common tick in Central Europe [41]. However, data on *Bartonella* species in rodents and their parasitizing ectoparasites, such as *C. nobilis* and *I. ricinus*, is lacking in Germany and information about host-vector-pathogen associations of the latter are scarce in Europe. Thus, this study's objectives were to analyse (1) the occurrence of *Bartonella* species in rodents, their fleas and ticks and (2) to identify their host-vector-pathogen associations in Germany.

Materials and Methods

Study Sites

The study site was divided in five animal trapping areas located around the city centre of Leipzig in Saxony which were formerly consecutively named from E to I [45]. Sites E–G are part of the "Leipziger Neuseenland" (http://www.leipzigerneuseenland.de/), a recultivated former brown coal mining area which is surrounding a lake (Lake Cospuden, 436 ha) and which belongs to the largest recreational area near Leipzig. Site H is a recreational city park called "Park Lößnig-Dölitz" (http://www.leipzig.de/de/buerger/freizeit/leipzig/parks/park_loessnig/), near Leipzig which contains several ponds. This park merges into forested areas and is located near a renatured waste disposal mountain. Site I (42 ha) is a game park, located in Leipzig, often frequented by visitors, which belongs to one of the largest riparian forests in Germany (www.leipziger-auwald.de). About 250 game animals are present (http://www.leipzig.de/de/buerger/freizeit/leipzig/parks/wildpark/wissen/). The study site and trapping areas have been described in detail in previous studies [45].

Sampling of Small Mammals

Small mammals were trapped with Sherman© live animal traps (H. B. Sherman Traps, Inc., Tallahassee, FL, USA) in 2010 and 2011 (official trapping permit: 36.11-36.45.12/4/10-026-MH, 364.60/2009-102-2). Traps, baited with apple slices, were placed for at least two consecutive nights per month and site and were checked twice a day. Small mammals were captured with at least 20 traps per subdivided area per month in August and October in 2010, from March to June in 2011 (sites E, H, I) and further in November 2011. Collected animals were anaesthetized with CO_2, euthanized by cervical dislocation and then stored at −80 °C until further examinations in the laboratory. Details about small mammal trapping were published before [30]. All small mammals were morphologically identified [48]. Bycaught small mammal species, found dead in traps or brought by a cat, were additionally verified by the use of a conventional PCR targeting the partial *cytochrome b* gene (354 bp) as described previously [34, 45]. Necropsy was carried out and spleens from the small mammals were collected.

Collection of Ticks and Fleas

Ectoparasites were collected from the trapped small mammals with tweezers and stored separately for each small mammal in 1.5-ml-sized tubes at −80 °C (Eppendorf, Hamburg, Germany) until further examination. The morphological identification of fleas and ticks was performed under a stereomicroscope with standardised taxonomic keys [17, 36, 37, 47]. Fleas and ticks were kept individually in separate 2.0-ml-sized tubes (Eppendorf) for the following DNA extraction.

DNA Extraction

DNA samples of spleens ($n = 79$) and ticks ($n = 365$) from the small mammals were available from a previous study on tick-borne infectious agents [45]. DNA extraction from collected fleas was carried out for each flea individually with the QIAamp DNA Mini Kit (Qiagen, Hilden, Germany) as previously described [46]. Quantity and quality of the extracted DNA were determined with a spectrophotometer (Nano Drop ND-1000, Erlangen, Germany).

Polymerase Chain Reaction

A conventional PCR targeting 635 bp of the *16S–23S* ITS region was performed for detection of *Bartonella* spp. as

described [25, 43], followed by UV light analysis on 2 % agarose gel dyed with GelRed (Biotium, Hayward CA, USA).

Sequence Analysis

PCR products were purified with the QIAquick PCR product purification Kit (Qiagen) according to the manufacturer's instructions and sequenced commercially by Eurofins MWG Operon (Martinsried, Germany). Sequences were analysed with Chromas Lite® (Technelysium Pty Ltd., South Brisbane, Australia) and compared with sequences obtained from GenBank using BLASTn (National Center for Biotechnology Information, Bethesda MD, USA) as previously described [43]. Analysed and aligned sequences were deposited in GenBank under the following Acc. Nos.: KU886369–KU886524.

Statistical Analysis

Confidence intervals (95 %CI) for prevalences of *Bartonella* spp. in small mammals, ticks and fleas were determined by the Clopper and Pearson method with GraphPad Software (GraphPad Software Inc., San Diego, CA, USA). Independence of compared small sample sizes ($n<30$) was tested with Fisher's exact test. The Kruskal-Wallis test with a Bonferroni correction and a significance level of $\alpha=0.05$ was used to test the independence of flea infestation on rodents per study site. The *t* test was used to test the independence of flea infestation on *M. glareolus* and *Apodemus* spp.

Results

Small Mammal, Flea and Tick Collections

Small Mammals

n total, 79 spleen DNA samples were available from small mammals belonging to seven different species (36 *M. glareolus*, 32 *Apodemus flavicollis*, 3 *Apodemus agrarius*, 3 *Arvicola terrestris*, 3 *Crocidura russula*, 1 *Sorex araneus*, 1 *Talpa europaea*). Altogether, 22 individuals were captured at site I, 30 at site H and 27 at sites E–G. Distribution of each small mammal species per trapping site was described before [45].

Fleas

A total of 135 fleas were collected from 42 rodents of the species *A. flavicollis*, *M. glareolus* and *A. agrarius*. The three *Ar. terrestris* and the five insectivores included in this study were not infested by fleas. The flea species were identified as *C. nobilis* ($n=67$; 34 females, 33 males), *Megabothris turbidus* ($n=40$; 28 females, 11 males, 1 unknown), *Nosopsyllus fasciatus* ($n=26$; 17 females, 9 males), *Doratopsylla dasycnema* ($n=1$; 1 unknown) and *Megabothris walkeri* ($n=1$; 1 male). Overall, the infestation rate of fleas was significantly higher on rodents belonging to the genus *Apodemus* compared to *M. glareolus* ($p<0.0001$; $t=367.5$). Significantly more *C. nobilis* were collected from *A. flavicollis* and *A. agrarius* (62.7 %) compared to *M. glareolus* (34.3 %) ($p<0.0001$; $t=4.8005$). Significantly more *Me. turbidus* were collected from *M. glareolus* (70 %) compared to *Apodemus* spp. (30 %) ($p=0.0001$; $t=4.0519$). In contrast, significantly more *N. fasciatus* (84.6 %) were collected from *A. flavicollis* (65.4 %) and *A. agrarius* (19.2 %) while only 15.4 % were collected from *M. glareolus* ($p=0.0004$; $t=3.7174$).

Details about flea infestation rates per rodent species are shown in Table 1. Altogether, 15 rodents were infested with more than one flea species: seven *M. glareolus* were infested with *Me. turbidus* and *C. nobilis*, one *M. glareolus* with *Me. turbidus*, *N. fasciatus* and *C. nobilis*; one *M. glareolus* with *Me. turbidus*, *Me. walkeri* and *C. nobilis*; two *A. flavicollis* with *Me. turbidus* and *C. nobilis*; three *A. flavicollis* with *N. fasciatus* and *C. nobilis* and one *A. flavicollis* with *Me. turbidus*, *N. fasciatus* and *C. nobilis*.

The infestation rate of fleas on rodents per site differed significantly ($\chi^2=9.038$; $p=0.0109$) as most fleas per rodent (4.2) were detected at site H whereas 1.3 fleas per rodent were detected at site I and 2.9 at the sites E–G. Moreover, the highest density of captured rodents was detected at site H ($n=30$) compared to site I ($n=22$) and sites E–G ($n=27$). Regarding seasonal dynamics, most fleas per rodent were collected in June (4.4) and the least in March (1.0). Altogether, 2.6 fleas per rodent were detected in August and 2.8 in October. The unequal distribution of collected rodents per month does not allow a statistical comparison between months for the flea infestation per rodent. As two *C. nobilis* were gone missing, 133 of the 135 collected fleas were further processed.

Ticks

Altogether, 365 DNA samples from attached ticks belonging to three species were available for this study (1 *I. trianguliceps*; 232 *I. ricinus*; 13 *Ixodes* spp.; 117 *Dermacentor reticulatus*; 2 *Dermacentor* spp.). *I. ricinus* ticks were mainly collected from *A. flavicollis* (50.6 %) and *M. glareolus* (*33.7*%) and from the other small mammal species: *S. araneus* (6.7 %), *Ar. terrestris* (7.8 %) and *A. agrarius* (1.2 %). In contrast, significantly more *D. reticulatus* were collected from *M. glareolus* (99.2 %) than from *A. flavicollis* (0.8 %) ($p=0.0045$; $t=2.9385$).

Table 1 Number of collected fleas per rodent species

Small mammal species	Total no.	Total no. of rodents infested by fleas (%)	Fleas species No. of rodents infested/total no. of fleas (range; median)									
			Ctenophthalmus nobilis		*Megabothris turbidus*		*Nosopsyllus fasciatus*		*Doratopsylla dasycnema*		*Megabothris walkeri*	
Apodemus flavicollis	32	20 (62.5)	12	40 (1–3; 2)	8	12 (1–4; 1)	6	17 (1–10; 1.5)	1	1		
Myodes glareolus	36	20 (55.6)	14	25 (1–4; 1.5)	12	28 (1–5; 2)	3	4 (1–2; 1)			1	1
Apodemus agrarius	3	2 (66.7)	1	2			1	5				
Total	71	42	27	67	20	40	10	26	1	1	1	1

The insectivores ($n = 5$) included in this study were not infested by fleas

Table 2 shows the numbers and species of ticks collected from small mammals. Further information about collected ticks was published before [45].

PCR Analysis for *Bartonella* spp.

Bartonella spp. in Rodents

In total, 52 of 79 (65.8 %; 95 % CI, 54.8–75.4) small mammals contained DNA of *Bartonella* spp. For *A. flavicollis*, 27 of 32 (84.4 %; 95 % CI, 67.8–93.6) and for *M. glareolus*, 21 of 36 (52.8 %; 95 % CI, 37.0–68.0) individuals were positive. Though not being infested with fleas, captured *Ar. terrestris* ($n = 3$; 100 %; 95 % CI, 38.1–100) and *T. europaea* ($n = 1$; 100 %; 95 % CI, 16.8–100) were positive for *Bartonella* spp. *S. araneus* ($n = 1$), *Cr. russula* ($n = 3$) and *A. agrarius* ($n = 3$) were negative (Table 3). Significantly more *A. flavicollis* were positive for *Bartonella* spp. compared to *M. glareolus* ($p = 0.0088$). Comparing sites, significantly more small mammals were found positive at site I (19 of 22) than at the other sites (E–G, 14 of 27; H, 18 of 30) ($p = 0.0341$).

Sequence Analysis for Bartonella spp. in Rodents

All PCR-positive samples were sequenced of which 29 uncultured *Bartonella* sp. showed 100 % homology to different sequences deposited in GenBank [Acc. Nos.: AJ269796 ($n = 5$, host: *Apodemus sylvaticus*, country: UK), DQ155380 ($n = 5$ host: *M. glareolus*, country: UK), DQ155387 ($n = 5$ host: *A. flavicollis*, country: Slovenia), DQ155391 ($n = 4$ host: *A. flavicollis*, country: Slovenia), DQ155376 ($n = 3$ host: *A. agrarius*, country: Slovenia), DQ155381 ($n = 2$ host: *M. glareolus*, country: Slovenia), DQ155379 ($n = 2$ host: *M. glareolus*, country: Slovenia), KP010210 ($n = 1$ host: *Eidolon dupreanum*, country: Madagascar), JX276643 ($n = 1$ host: *A. sylvaticus*, country: Spain), JN403036 ($n = 1$ host: *A. sylvaticus*, country: Georgia)]. These different sequences from GenBank showed identity of 22–96 % compared to each other. Overall, sequences showing 100 % identity to *Bartonella grahamii* ($n = 12$; CP001562), 96–100 % identity to *B. taylorii* [AJ269788 ($n = 4$), AJ269796 ($n = 2$), AJ269784 ($n = 1$), HE974357 ($n = 1$)] and 100 % identity to *Bartonella* sp. N40 ($n = 3$; AJ269791) were detected in small mammals. Three small mammals showed double infections: one

Table 2 Number of collected ticks per small mammal species

Small mammal species	Total no.	Total no. of rodents infested by ticks (%)	Tick species No. of rodents infested/total no. of ticks (range; median)									
			Ixodes ricinus		*Ixodes* spp.		*Dermacentor reticulatus*		*Dermacentor* spp.		*Ixodes trianguliceps*	
Apodemus flavicollis	32	20 (62.5)	24	116 (1–21; 9.5)	4	9 (1–2; 1.5)	1	1 (5)				
Myodes glareolus	36	21 (58.3)	17	79 (1–14; 6)	2	3 (1–2; 1.5)	13	116 (1–22; 10)	2	2 (1)	1	1
Apodemus agrarius	3	2 (66.7)	2	18 (1–17; 8.5)			1	5 (5)				
Arvicola terrestris	3	2 (66.7)	2	3 (1–2; 1.5)								
Soricidae	4	2 (50)	2	12 (1–11; 5.5)								
Talpa europaea	1	1 (100)	1	4 (4)								
Total	79	48	48	232	6	12	15	122	2	2	1	1

Table 3 Prevalences of uncultured *Bartonella* sp., *B. grahamii*, *B. taylorii* and *Bartonella* sp. N40 in spleen samples from different small mammal species

Small mammal species	Total no.	Total no. of small mammals infected with *Bartonella*[a] (%)	*Bartonella* species Total infected (%)			
			Uncultured *Bartonella* sp.	*B. grahamii*	*B. taylorii*	*Bartonella* sp. N40
Apodemus flavicollis	32	27 (84.4)	18 (62.0)	4 (33.3)	3 (37.5)	2 (66.7)
Myodes glareolus	36	21 (52.8)	9 (31.0)	6 (50)	5 (62.5)	1 (33.3)
Apodemus agrarius	3	0 (0)				
Arvicola terrestris	3	3 (100)	1 (3.5)	2 (16.7)		
Crocidura russula	3	0 (0)				
Sorex araneus	1	0 (0)				
Talpa europaea	1	1 (100)	1 (3.5)			
Total	79	52 (65.8)	29 (100)	12 (100)	8 (100)	3 (100)

[a] Numbers differ from sum of individual *Bartonella* infections, because of double infections in individual animals (1 *M. glareolus*: B. uncultured + *B. taylorii*, 1 *M. glareolus*: uncultured *Bartonella* sp. + *B. grahamii*; 1 *A. flavicollis*: uncultured *Bartonella* sp. + uncultured *Bartonella* sp.)

A. flavicollis with two different uncultured *Bartonella* sp. (Acc. Nos.: AJ269796 and JN403036, showing identity of 50.3 % compared to each other), one *M. glareolus* with uncultured *Bartonella* sp. and *B. grahamii* and one *M. glareolus* with uncultured *Bartonella* sp. and *B. taylorii* (Table 3). Additionally, sequence analysis revealed one sequence in *A. flavicollis* which showed low identity of 71 % to several different *Bartonella* spp. sequences deposited in GenBank (Table 4). Moreover, amplicons detected by conventional PCR in two *A. flavicollis* could not be further determined by sequencing.

M. glareolus were mainly infected by *B. grahamii* and *B. taylorii* (50 and 62.5 %, respectively) while uncultured *Bartonella* sp. and *Bartonella* sp. N40 were mainly detected in *A. flavicollis* (62 and 66.7 %, respectively).

Bartonella spp. in Fleas

A total of 54.1 % (73/133) of the fleas were positive for DNA of *Bartonella* spp. (95 % CI, 45.7–62.4). In total, 29 of 65 *C. nobilis*, 17 of 26 *N. fasciatus* and 26 of 40 *Me. turbidus* were positive. While the single *D. dasycnema* was positive, the single *Me. walkeri* was negative for *Bartonella* spp. Comparing prevalences at sites, ten of 13 (76.9 %) at site I, 28 of 53 (52.8 %) at sites E–G and 35 of 67 (52.2 %) fleas at site H were positive for *Bartonella* spp. The prevalences did not differ significantly ($\chi^2 = 2.93$; $p = 0.2305$). Positive fleas were collected from 23 positive and 12 negative rodents. Altogether, 15 of the 23 positive rodents were positive for the same *Bartonella* sp. as their fleas. Positive fleas which were collected from the 12 negative small mammals were positive for uncultured *Bartonella* sp. ($n = 13$ identity of 97–100 % to the following GenBank Acc. Nos.: DQ155387 ($n = 2$), DQ155391 ($n = 1$), DQ155380 ($n = 3$), JN403036 ($n = 5$) and AY877424 ($n = 2$, host: *Stenoponia tripectinata*, country: Portugal)), *B. grahamii* ($n = 2$; identity to the following Acc. No.: JF766266), *B. taylorii* ($n = 2$; identity to the following Acc. No.: JF766266), *Bartonella* sp. N40 ($n = 1$; identity to the following Acc. No.: JF766266) and *B. elizabethae* ($n = 1$; identity to the following Acc. No.: JF766266).

Sequencing of Bartonella spp. in Fleas

Sequencing was performed for all positive flea samples ($n = 73$) (Table 4). Altogether, 96–100 % identity was detected for uncultured *Bartonella* sp. [$n = 45$; GenBank Acc. Nos.: DQ155387 ($n = 12$), DQ155380 ($n = 11$), DQ155391 ($n = 8$), JN403036 ($n = 8$), AY877424 ($n = 3$), DQ155384 ($n = 1$), HM596446 ($n = 1$), AJ269794 ($n = 1$)]. Overall, 14 sequences showed 100 % identity to *B. grahamii* [Acc. Nos.: CP001562 ($n = 13$), JN810855 ($n = 1$)] and eight sequences were 100 % identical to *B. taylorii* [Acc. Nos.: AJ269788 ($n = 6$), AJ269784 ($n = 2$)]. Sequences were 100 % identical to *B.* sp. N40 (Acc. No.: AJ269787 ($n = 4$)) and *B. elizabethae* (Acc. No.: JF766266 ($n = 2$)). Double infections of uncultured *Bartonella* sp. (Acc. No.: JN403036, $n = 2$) and *B. elizabethae* were detected in one *N. fasciatus* and one *C. nobilis*. Sequencing of amplicons from two *C. nobilis*, two *Me. turbidus* and three *N. fasciatus* did not succeed.

B. grahamii was detected significantly more often in *Me. turbidus* (93 %; $\chi^2 = 29.6255$; $p < 0.0001$) than in *N. fasciatus* (7 %) and *C. nobilis* (0 %). *Bartonella* sp. N40 and *B. taylorii* were mainly but not significantly more often detected in *C. nobilis* (80 % respectively 50 %, $p = 0.2945$). Uncultured *Bartonella* sp. was mainly detected in *N. fasciatus* (28.9 %) and *C. nobilis* (46.7 %) however not significantly ($p = 0.1325$)

Table 4 Prevalences of uncultured *Bartonella* sp., *B. grahamii*, *B. taylorii*, *Bartonella* sp. N40 and *B. elizabethae* in fleas collected from rodents

Flea species	Total no.	Total no. of fleas infected with *Bartonella*[a] (%)	*Bartonella* species Total infected (%)				
			Uncultured *Bartonella* sp.	*B. grahamii*	*B. taylorii*	*Bartonella* sp. N40	*B. elizabethae*
Ctenophthalmus nobilis	67	29 (43.3)	21 (46.7)		4 (50)	3 (75)	1 (50)
Megabothris turbidus	40	26 (62.5)	10 (22.2)	13 (92.9)	2 (25)		
Nosopsyllus fasciatus	26	17 (92.9)	13 (28.9)	1 (7.1)	2 (25)	1 (25)	1 (50)
Doratopsylla dasycnema	1	1 (100)	1 (2.2)				
Megabothris walkeri	1	0 (0)					
Total	135	73 (53.3)	45 (100)	14 (100)	8 (100)	4 (100)	2 (100)

[a] Numbers differ from sum of individual *Bartonella* infections, because of double infections in individual animals (1 *C. nobilis*: uncultured *Bartonella* sp. + *B. elizabethae*, 1 *N. fasciatus*: uncultured *Bartonella* sp. + *B. elizabethae*)

compared to *Me. turbidus*. *B. elizabethae* was exclusively detected in fleas.

Bartonella spp. in Ticks

Altogether, 51 of the 365 ticks were positive for *Bartonella* spp. (16.3 %; 95 % CI, 12.9–20.5). In total, 39 of 232 *I. ricinus* (19.1 %; 95 % CI, 14.4–24.6) and 6 of 117 *D. reticulatus* (5.9 %; 95 % CI, 3.9–14.1) were positive for *Bartonella* spp. (Table 5). Six of 13 *Ixodes* spp. which could not be further determined were tested positive for *Bartonella* spp. (50.0 %; 95 % CI, 21.3–72.0). *Dermacentor s*pp. were *Bartonella* negative ($n=2$; 95 % CI, 0–70.1) as well as *I. trianguliceps* ($n=1$; 95 % CI, 0–83.3). *I. ricinus* ticks were significantly more often infected with *Bartonella* spp. than *D. reticulatus* ($p=0.002$). The prevalence at the sites E–G (10 of 194) was significantly lower ($\chi^2=31.3108$; $p<0.0001$) than at both other sites H (30 of 109) and I (11 of 62). Positive ticks were collected from 19 positive and three negative small mammals. For 18 of the 19 positive rodents, *Bartonella* sequences were matching those from their ticks. One *B. taylorii*-positive rodent was infested with a *B. grahamii*-positive tick.

Sequence Analysis for Bartonella spp. in Ticks

Sequencing was performed for 35 selected tick samples. Altogether, 20 uncultured *Bartonella* sp. [96–100 %; identity to the following Acc. Nos.: DQ155387 ($n=7$), AJ269787 ($n=4$), DQ155380 ($n=3$), DQ155374 ($n=1$), DQ155376 ($n=1$), DQ155379 ($n=1$), DQ155381 ($n=1$), DQ155386 ($n=1$)], 10 *B. grahamii* [identity to Acc. Nos.: CP001562 ($n=9$), KF774307 ($n=1$)], three *Bartonella chomelii* (identity to Acc. No.: KM215715 ($n=3$)] and two *B. taylorii* [identity of 99–100 % to Acc. Nos.: JN366404 ($n=1$), AJ269788 ($n=1$)] were detected in ticks (Table 5). Double infections were not detected. Overall, 16 tick samples, detected positive by conventional PCR, were not further processed by sequencing due to previous selection criteria. Additionally, PCR products of five *I. ricinus*, two *D. reticulatus* and one *Dermacentor* spp. could not be determined by sequencing.

Uncultured *Bartonella* spp. were significantly more often detected in *I. ricinus* than in *D. reticulatus* (90.0 %; $\chi^2=6.8997$; $p=0.0086$).

Host-Vector-Pathogen Associations

The complexity of relations between *Bartonella* spp., flea and rodent species is shown in Fig. 1. In both fleas and in rodents, uncultured *Bartonella* spp. were most commonly detected. The figures shows clearly that Bartonella spp. detected in most flea and rodent species are very diverse and that there is just a single host-vector-pathogen association between *M. glareolus*, *Me. turbidus* and *B. grahamii*.

Discussion

In comparison to other mammals, the highest diversity of *Bartonella* spp. was described in rodents to date. Altogether, 22 rodent-associated *Bartonella* spp. and subspecies were described [14]. Some of them were initially isolated from infected humans [8]. The prevalences for *Bartonella* spp. in rodents from other European countries such as Poland, Denmark, England and Sweden ranged between 11 and 72 % [14]. Results of our study (65.8 %) therefore fall into the upper range of those studies. Co-infections of hosts and vectors with multiple *Bartonella* species or genotypes are a familiar phenomenon [14]. In the present study, several rodents were also co-infected with two different *Bartonella* spp. Most commonly, different uncultured *Bartonella* spp. were detected in

Table 5 Prevalences of uncultured *Bartonella* sp., *B. grahamii*, *B. taylorii* and *B. chomelii* in ticks collected from different small mammal species

Tick species	Total no.	Total no. of ticks infected with *Bartonella*[a] (%)	*Bartonella* species Total infected (%)				
			Uncultured *Bartonella* sp.	*B. grahamii*	*B. taylorii*	*Bartonella* sp. N40	*B. chomelii*
Ixodes ricinus	232	39 (17.0)	17 (90.0)	6 (60.0)	1 (50.0)		2 (66.7)
Dermacentor reticulatus	117	6 (5.2)	1 (5.0)	4 (40.0)			1 (33.3)
Ixodes spp.	13	6 (46.2)	1 (5.0)		1 (50.0)	1 (100)	
Dermacentor spp.	2	0 (0)					
Ixodes trianguliceps	1	0 (0)					
Total	365	51 (14.2)	19 (100)	10 (100)	2 (100)	1 (100)	3 (100)

[a] Numbers differ from sum of individual *Bartonella* infections, because 16 of all 51 positive samples were not further determined by sequencing

combination with *B. grahamii* and *B. taylorii* which are common *Bartonella* species in rodent hosts [14]. In this study, mostly uncultured Bartonella spp. were found in rodents. In a recent study from France, a high prevalence for *Bartonella* spp. was detected in bank voles (39.3 %). However, focusing on the *Bartonella* species characterisation, also yet unknown genotypes were found as in our study [8]. *A. flavicollis* was significantly more often infected with *Bartonella* spp. than *M. glareolus* in the present study. This observation was made before and it was not explained by the different ecologic habitats of both of these rodent species but rather by specific host-pathogen interactions. While *M. glareolus* is able to resolve *Bartonella* infections within 1–2 months, the duration of resolving the infection is still not definitely known for *A. flavicollis* [35]. Moreover, it has been observed that reinfection in *M. glareolus* is more seldom than in *A. flavicollis* [35]. The flea abundance might directly influence the prevalence of *Bartonella* infection in the rodent host. Overall, fleas were found significantly more often on *Apodemus* spp. than on *M. glareolus* in this study. This may also explain why *Apodemus* spp. were more often infected with *Bartonella* sp. than *M. glareolus*. Furthermore, *M. glareolus*, in contrast to other small mammal species such as *Peromyscus leucopus*, is not able to transmit *Bartonella* spp. transplacentally [5, 23]. This is why vectors are important for the maintenance of *Bartonella* in these reservoir hosts. *B. grahamii* was more abundant in *M. glareolus* than in *Apodemus* spp. in our study. Previously, it has been reported that *M. glareolus* is the main

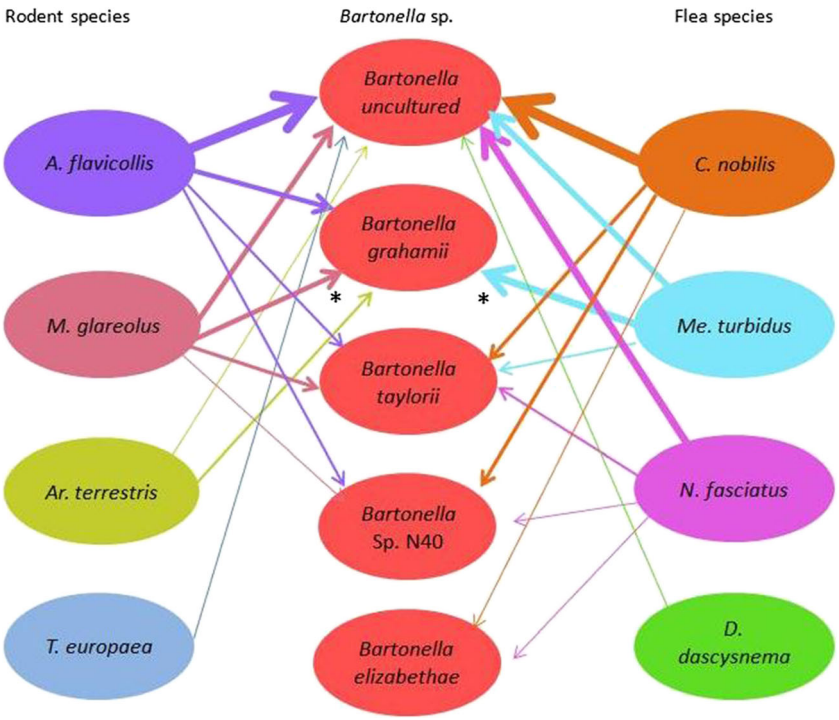


Fig. 1 Complex associations between *Bartonella*, rodent and flea species. *C. nobilis* = *Ctenophthalmus nobilis*, *N. fasciatus* = *Nosopsyllus fasciatus*, *D. dasycnema* = *Doratopsylla dasycnema*, *Me. turbidus* = *Megabothris turbidus*, *A. flavicollis* = *Apodemus flavicollis*, *M. glareolus* = *Myodes glareolus*, *T. europaea* = *Talpa europaea*, *Ar. terrestris* = *Arvicola terrestris*. *Statistically significant

reservoir for *B. grahamii* [53]. A former study reported that *B. grahamii* may be transmitted to the paws of cats by infected rodents which were killed by cats, as contact samples of cat paws were positive [31]. In this way, this pathogenic *Bartonella* species may be further transmitted by cat scratches to other animals and humans [31].

T. europaea and *Ar. terrestris* were, though not infested with fleas, positive for *Bartonella* spp. in our study. It is probable that those small mammals, which were found dead due to being caught by a cat, were not infested with fleas due to the survival strategy of fleas which jump off dead hosts as soon as the hosts' body temperature is decreasing [18]. The high infestation rate of fleas on small mammals at site "I" could be due to the high density of captured rodents and may explain why the majority of *Bartonella*-positive rodents were found there. Fleas vary greatly in their degree of host specificity, from being highly host-specific to host-opportunistic [24]. Most flea species showed a preference for certain rodent species in our study. As *Me. turbidus* was found predominantly on *M. glareolus* and *C. nobilis* and *N. fasciatus* significantly more often on *Apodemus* spp., it seems likely that these flea species have a host preference, but no clear host specificity.

In our study, the prevalences for *Bartonella* spp. were high in most rodent flea species such as *C. nobilis* (43.3 %), *N. fasciatus* (92.9 %) and *Me. turbidus* (62.5 %). There are only few studies investigating those flea species for the occurrence of *Bartonella* spp. However, *B. taylorii* and *B. grahamii* are known to be transmitted by *C. nobilis* to *M. glareolus* [5]. *Bartonella* sp. has previously been described in *N. fasciatus* [33]. The prevalence of *Bartonella* spp. in fleas seemed to be independent of the *Bartonella* species and of the prevalence in their rodent hosts as most positive fleas derived either from negative rodents or from rodents which were not positive for the same *Bartonella* species. This observation has been made before, with the conclusion of a vertical transmission path for *Bartonella* in fleas [7]. *B. grahamii*, *B. taylorii* and *B. elizabethae* are known to occur in fleas which infest rodents [53]. In our study, these *Bartonella* spp. as well as several uncultured *Bartonella* spp. were found in fleas. Though *B. elizabethae* is known to occur in rodents and fleas, we detected this *Bartonella* species exclusively in fleas [6].

B. grahamii was more abundant in *Me. turbidus* fleas compared to other flea species. In accordance with this finding, *B. grahamii* was mainly found in *M. glareolus*, the main host of *Me. turbidus*, suggesting a host-vector-pathogen association for *M. glareolus*, *Me. turbidus* and *B. grahamii*. Apart from this observation, no other significant host-vector-pathogen associations were found.

To date, it is not clear whether ticks are involved in the natural life cycle of *Bartonella* spp. Their epidemiological importance has been considered secondary [16, 28, 50]. However, previous studies from Italy, France, Portugal and Germany reported prevalences ranging between 1.5 and 32.3 % in questing *I. ricinus* ticks respectively in *I. ricinus* removed from humans [10, 15, 20, 42]. In our study, the prevalence is in the range of the mentioned European studies (16.3 %). The *Bartonella* species detected in ticks in this study were *B. chomelii*, *B. grahamii*, *B. taylorii* and uncultured *Bartonella* sp. The prevalence in *I. ricinus* was significantly higher when compared to *D. reticulatus*. Thus, *D. reticulatus* appears to be less important in the transmission cycle of *Bartonella* spp. than *I. ricinus*. Comparing the results for *Bartonella* in ticks and fleas, ticks seem to be more dependent on the prevalence and *Bartonella* sp. detected in their rodent hosts in contrast to fleas. Most positive ticks were collected from positive rodents in our study, and the *Bartonella* sp. found in ticks were mainly matching the species found in their rodent hosts. These results suggest that ticks, reflecting the infection in their rodent hosts, are of minor importance in the natural life cycle of *Bartonella* spp. found in rodents and their fleas. However, *B. chomelii* was exclusively found in ticks but not in fleas and rodent hosts from this study. Formerly, *B. chomelii* was detected in domesticated ruminants and in their engorged ticks from Spain and Italy [1, 11]. Cattle are suggested to be reservoirs not showing any clinical symptoms [1]. As *B. chomelii* has never been detected in fleas and rodents thus far, this *Bartonella* sp. is likely to be transmitted transovarially in tick populations in nature as most positive ticks deriving from rodents were larvae which were naïve before feeding on their rodent hosts.

Conclusion

In conclusion, a high prevalence of different *Bartonella* species was found in rodents and their fleas with a high diversity and a tendency towards a host-vector-pathogen association for *M. glareolus*, *Me. turbidus* and *B. grahamii*. Zoonotic bartonellae such as *B. grahamii* and *B. elizabethae* were found in rodents and their fleas in this study. Therefore, humans and their companion animals are at risk of contracting *Bartonella* infection by direct or indirect contact to infected blood of wild living rodents. Ticks seem to be of minor importance in the transmission cycle of *Bartonella* sp. found in fleas and rodents. However, ticks may play an important role as vectors in the transmission of *B. chomelii*.

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